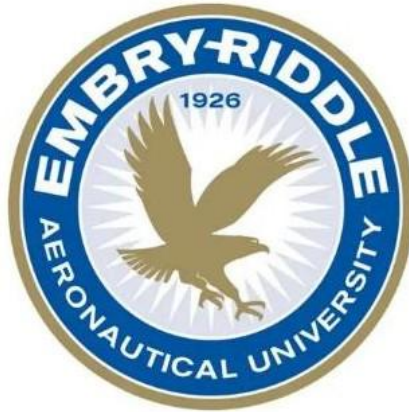


Servo Modeling



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Procedure

I. Bump Test Experiment

The main goal of this experiment is to find a transfer function that describes the gear system of the SRV02 servo-motor system. A step input was given to the SRV02, and the corresponding load shaft response was recorded to find the system's steady-state gain (K) and time constant (τ).

Experimental Procedure:

- Log into the computer and download the zip folder “Lab Session 01 - Exp 01 - Codes v2.zip”, provided on Canvas. Once downloaded, unzip the folder and save it in a directory/folder that can be easily accessible.
- Ensure the computer used has Simulink and MATLAB (2020 version) installed.
- Make sure the hardware provided consists of all the items required for the experiment. Refer to the “Quick Start Guide” section (Pg. 12) of the lab manual.
- Turn on the amplifier and connect the USB.
- Open and run “setup_srv02_exp01_mdl.m” file and open “q_srv02_mdl” file.
- In the “q_srv02_mdl” file, double-click on the Signal Generator block and ensure the following values:
 - Wave form: Square
 - Amplitude: 1.0
 - Frequency: 0.4
 - Units: Hertz
- Change the value of “spf src” block from 1 to 2.
- Set the Amplitude (V) slider to 1.5 V and the Offset (V) block to 2.0V.
- To observe how the system reacts, double-click on the load shaft speed scope, w_l (rad/s), and the motor input voltage scope, V_m (V).
- Set the Fixed-step size to 0.002 in the Solver details tab (Hardware Settings). The “Stop Time” can also be set to 5 sec instead of infinity.
- Click on “Monitor & Tune” to run the simulation. This should make the gears on the SRV02 rotate in the same direction and alternating between low and high speeds.
- The responses can be observed in the scopes. The data should be collected and available in the MATLAB workspace under the “Speed” variable.
- Save the data into a “.mat” file and plot. Then find the steady-state gain and time constant.

II. Model Validation Experiment

This experiment is performed after the Bump Test, and the goal is to adjust the model parameters found in the previous experiment to tune the transfer function. The objective is to match the simulated with the actual response of the system as closely as possible. In this process, the behavior of steady-state gain and time constant will be better understood.

Experimental Procedure:

- Since this experiment is performed right after the Bump Test, start by setting the Amplitude (V) slider to 1.0 V and the Offset (V) block to 1.5 V. Else, the bullets (1-7) from the experiment procedure of the Bump Test can be followed.
- Open the load shaft speed scope, w_l (rad/s), and the motor input voltage scope, V_m (V).
- Click on “Monitor & Tune” to run the simulation. The gears should start rotating.
- With the data collected, save the workshop data in a “.mat” file.
- Now, go to MATLAB Command Window and enter “K = 0.8” and “tau = 0.06”.
- Run the simulation by clicking “Monitor & Tune.”
- Save the collected data in a “.mat” file from MATLAB workspace.
- Enter “K = 1.25” and “tau = 0.2”. Run the simulation by clicking “Monitor & Tune.”
- Save the collected data in a “.mat” file from MATLAB workspace.
- Enter “K = 1.53” and “tau = 0.0253”. Run the simulation by clicking “Monitor & Tune.”
- Save the collected data in a “.mat” file from MATLAB workspace.
- With calculated “K” and “tau” values from the Bump Test, enter each value in the command window, respectively.
- Run the simulation and save the collected data.
- Iterate “K” and “tau” values until the simulated response matches the measured response.
- Save the collected data in a “.mat” file from MATLAB workspace.
- Make sure all required data is recorded. Turn off the power to the amplifier and disconnect the USB.

Results

Nominal Values:

The calculations for nominal values are illustrated in Appendix A.

$$K = 1.53 \frac{\text{rad/s}}{\text{V}}$$

$$\tau = 0.0253 \text{ s}$$

Bump Test Experiment:

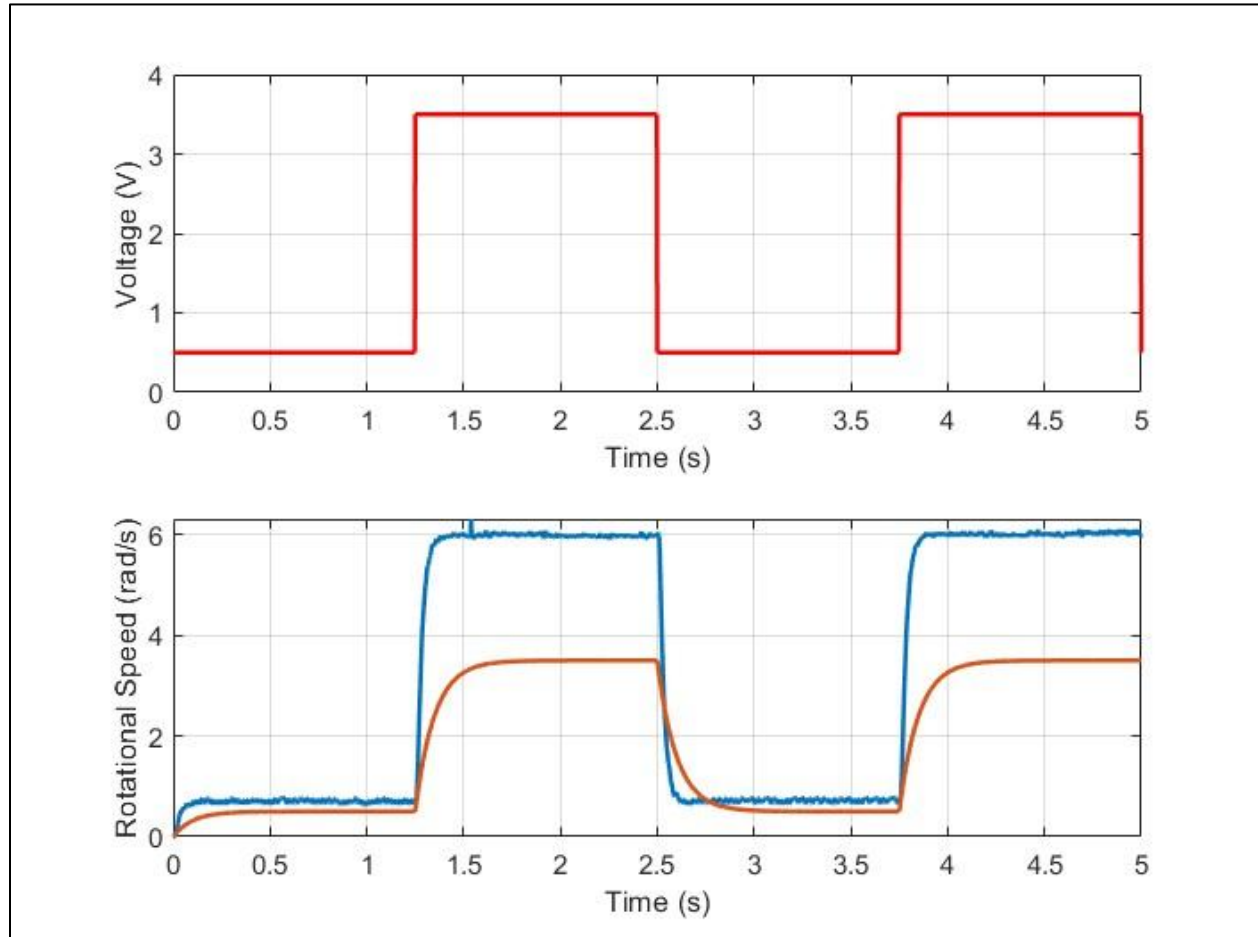


Figure 1: Voltage & Rotational Speed w.r.t Time

Model Validation Experiment:

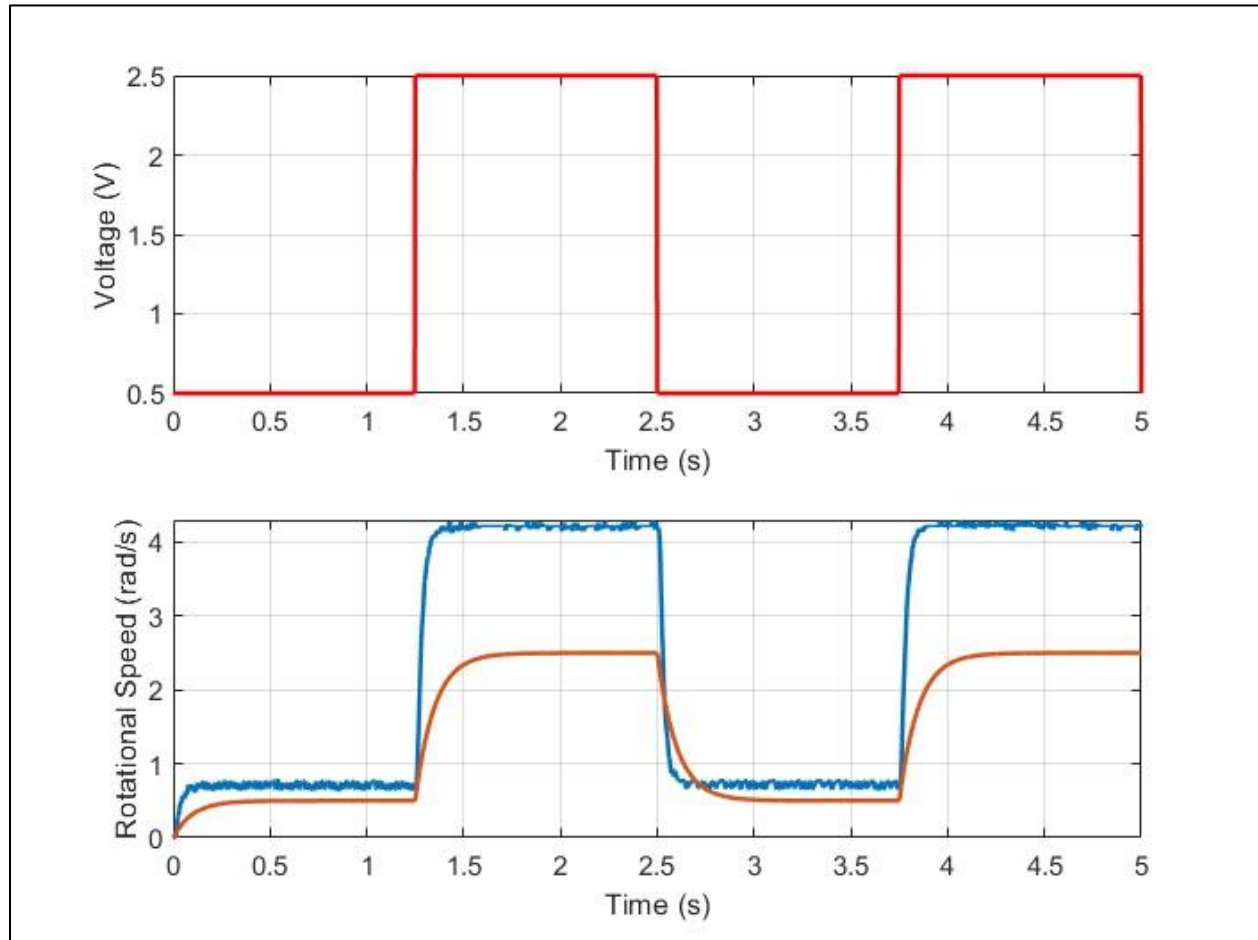


Figure 2: Voltage & Rotational Speed w.r.t Time.
 $K = 1 \text{ rad/s/V}$ and $\tau = 0.1 \text{ s}$.

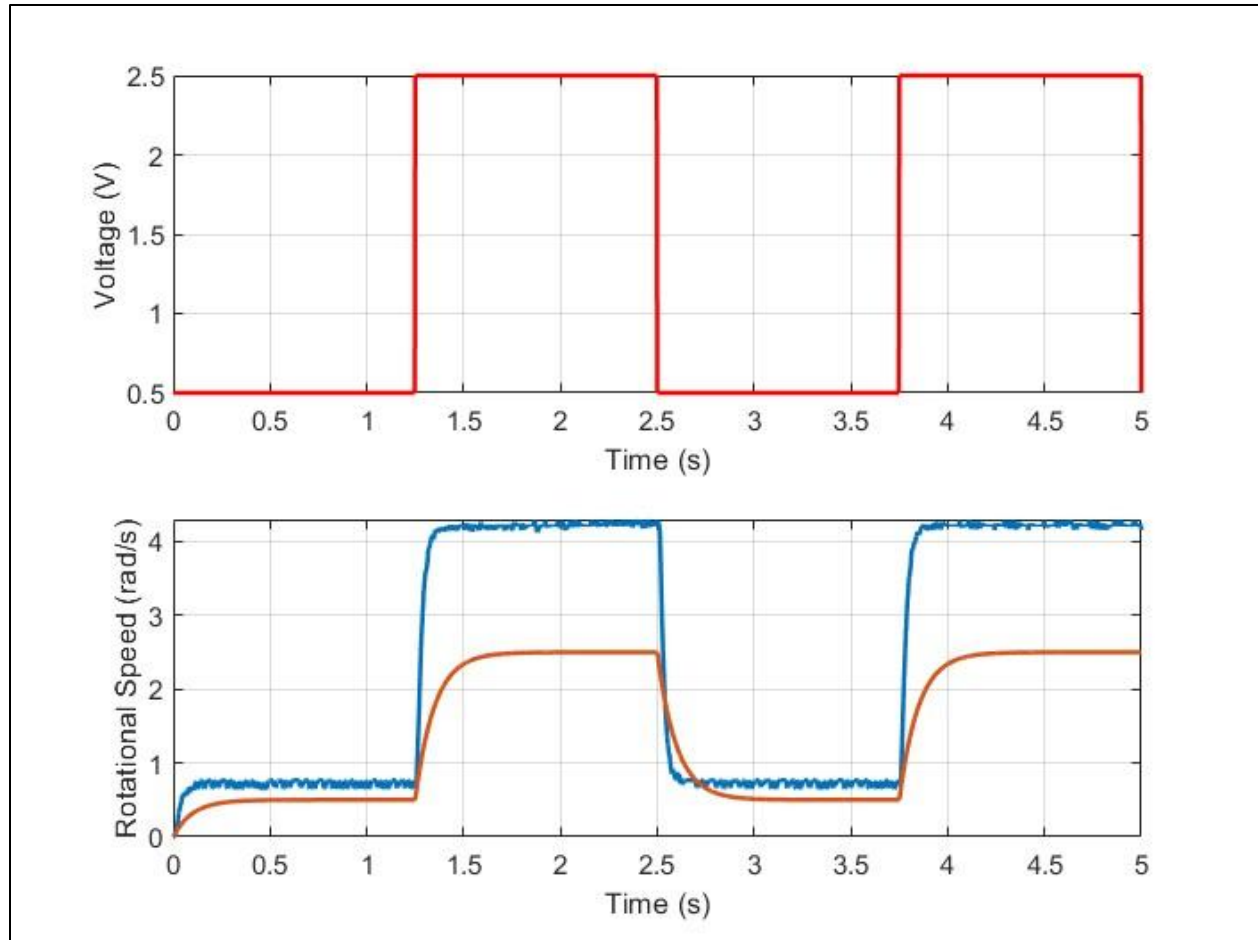


Figure 3: Voltage & Rotational Speed w.r.t Time.

$K = 0.8 \text{ rad/s/V}$ and $\tau = 0.06 \text{ s}$.

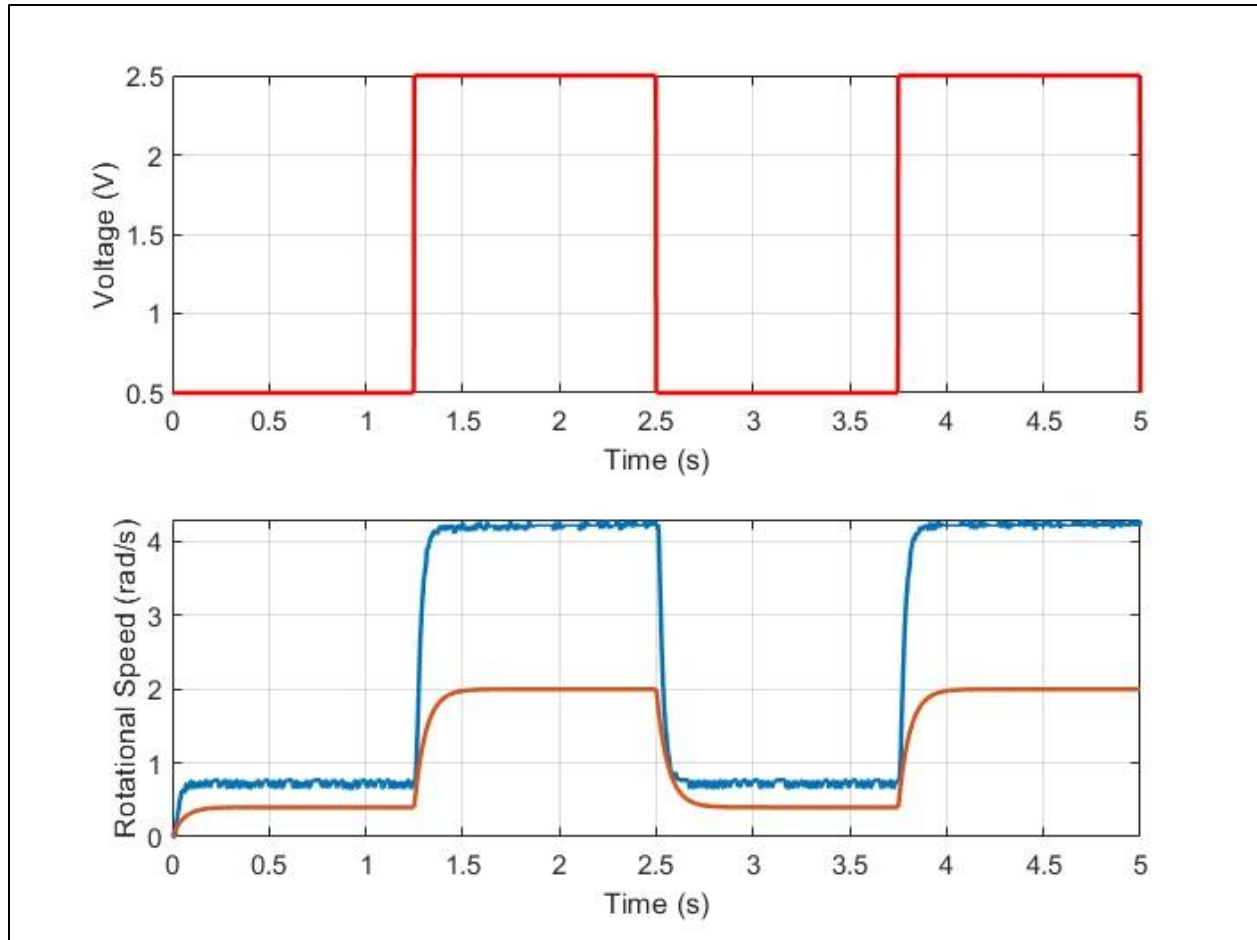


Figure 4: Voltage & Rotational Speed w.r.t Time.

$K = 1.25 \text{ rad/s/V}$ and $\tau = 0.2 \text{ s}$.

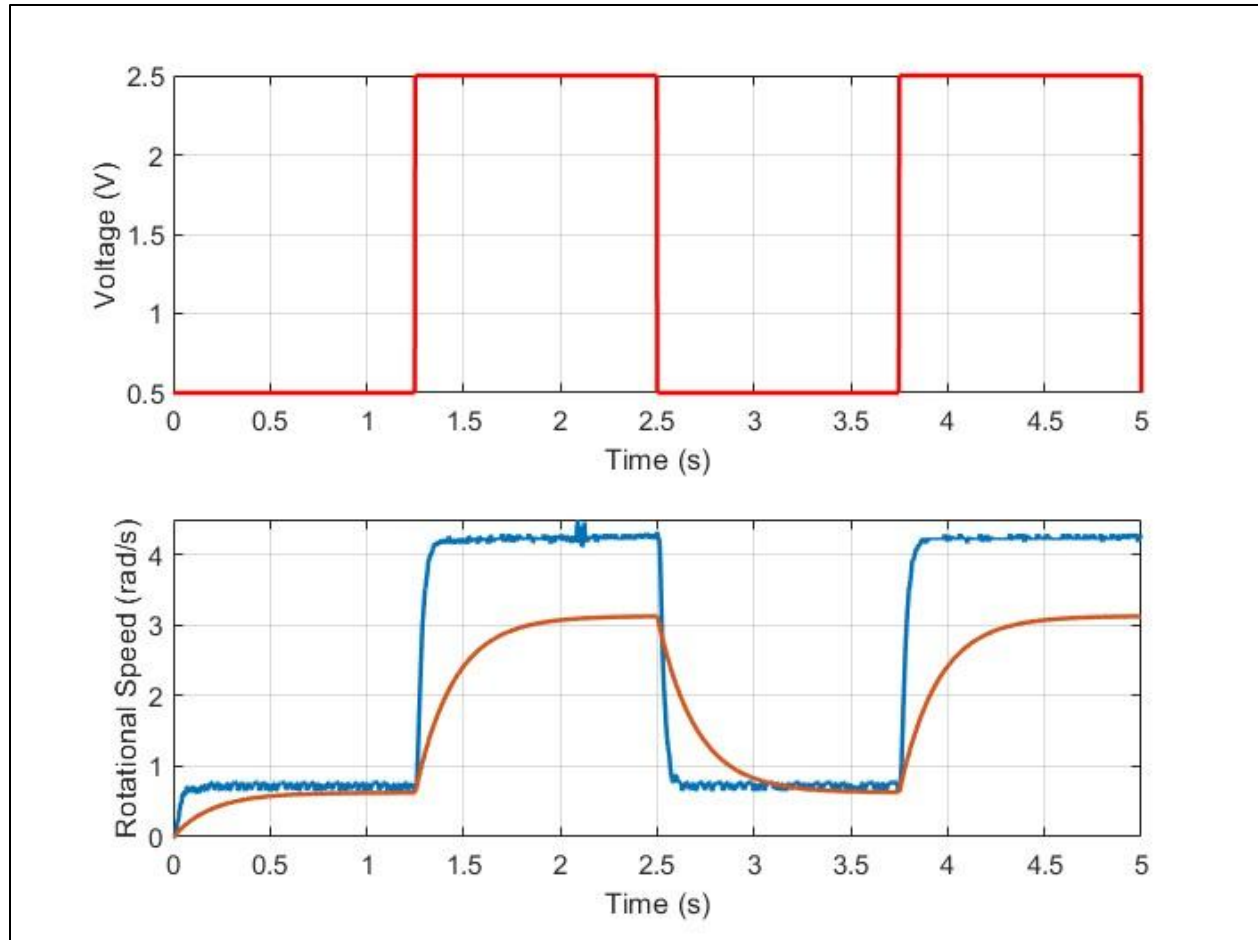


Figure 5: Voltage & Rotational Speed w.r.t Time.

$K = 1.53 \text{ rad/s/V}$ and $\tau = 0.0253 \text{ s}$.

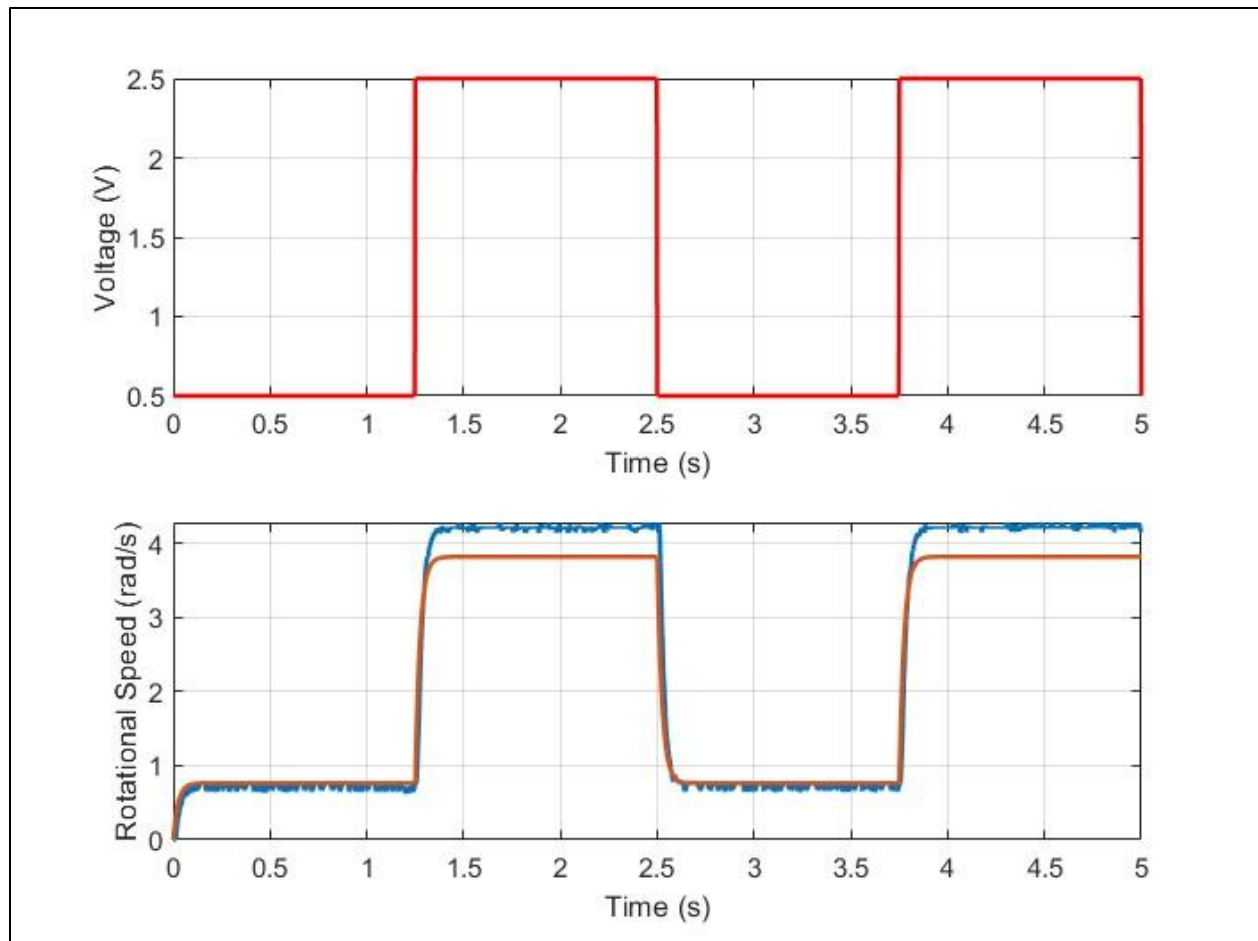


Figure 6: Voltage & Rotational Speed w.r.t Time.

$K = 1.7866 \text{ rad/s/V}$ and $\tau = 0.048 \text{ s}$.

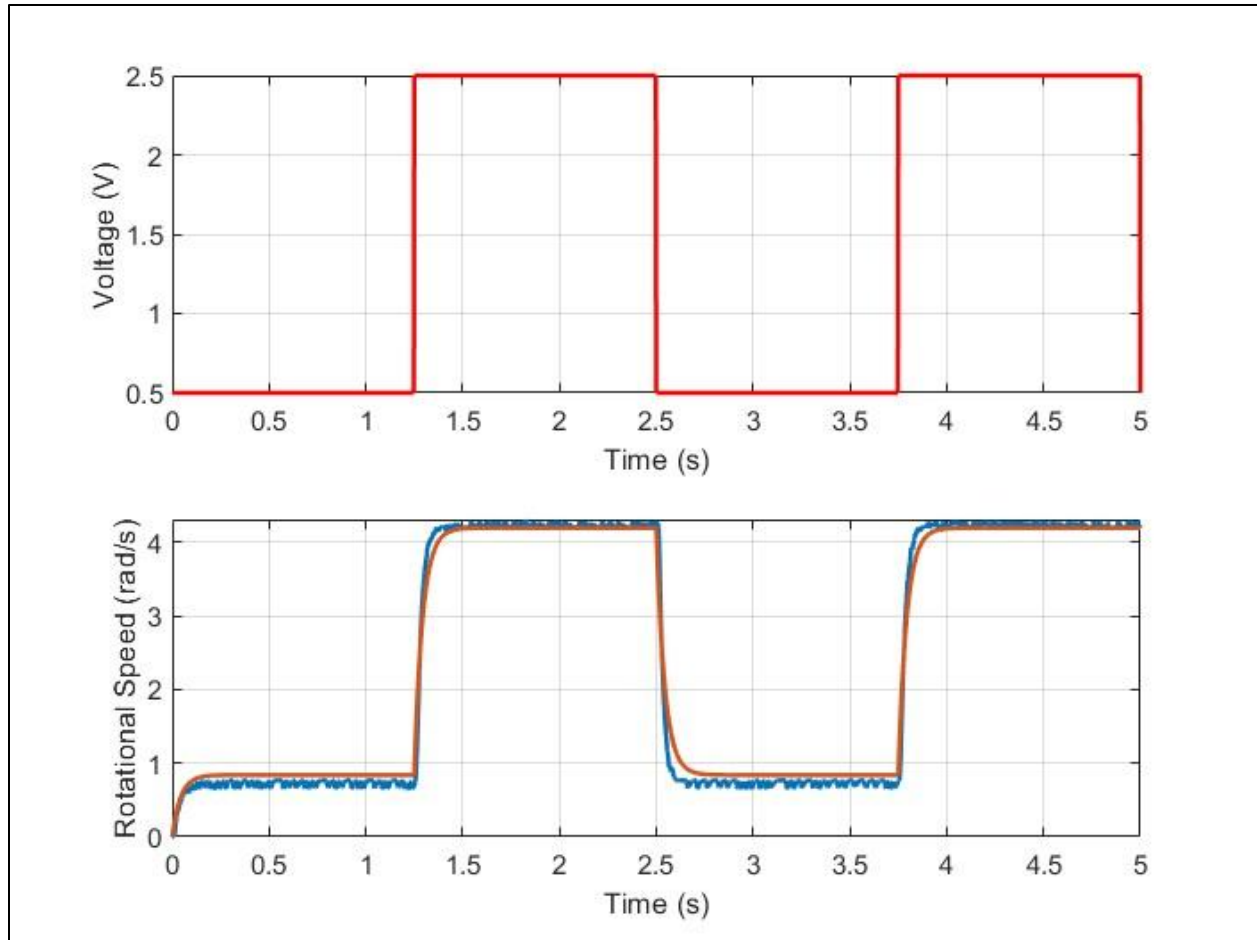


Figure 7: Voltage & Rotational Speed w.r.t Time.

$K = 1.68 \text{ rad/s/V}$ and $\tau = 0.044 \text{ s}$.

Description	Symbol	Value	Unit
Nominal Values			
Open-Loop Steady-State Gain	K	1.53	(rad/s)/V
Open-Loop Time Constant	τ	0.0253	s
Bump Test Exp.			
Open-Loop Steady-State Gain	$K_{e,b}$	1.7866	(rad/s)/V
Open-Loop Time Constant	$\tau_{e,b}$	0.048	s
Model Validation			
Open-Loop Steady-State Gain	$K_{e,v}$	1.68	(rad/s)/V
Open-Loop Time Constant	$\tau_{e,v}$	0.044	s

Table 1: Summary of Results for the SRV02 Modeling Laboratory

Analysis

To generate plots in Figure 6, the steady-state gain and the time constant need to be calculated from the Bump Test plot (Figure 1).

$$K_{e,b} = \frac{y_{ss} - y_0}{u_{max} - u_{min}}$$

With the help of MATLAB, the required graphs were plotted. Then, using the cursor and the coordinates shown in MATLAB, the required points were obtained. For instance, y_{ss} was obtained when the curve eventually settled at a steady-state value after the step was applied. y_0 is the initial value of the resulting output signal. u_{max} and u_{min} represent the minimum and maximum values of the input signal.

The following figure shows an example of the method used to obtain values from the plots produced in MATLAB.

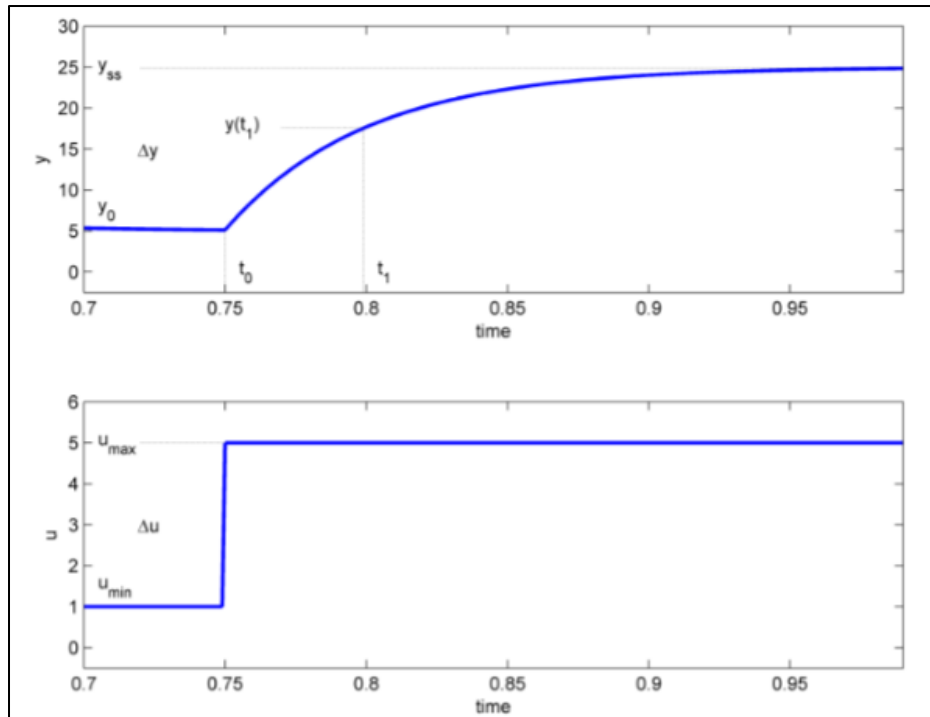


Figure 8: Input and Output Signal used in the Bump Test Method [1]

$$K_{e,b} = \frac{6.051 - 0.6912}{3.5 - 0.5} = 1.7866 \frac{\left(\frac{rad}{s}\right)}{V}$$

The next step is to find the time constant using the following equations:

$$\tau_{e,b} = t_1 - t_0$$

$$y(t_1) = 0.632 \times y_{ss} + y_0$$

For simplicity, drop the y_0 term.

$$y(t_1) = 0.632 \times y_{ss} = 0.632 \times 6.051 = 3.824$$

Now, using the plots in MATLAB, t_1 is found to be 1.25 s. Therefore, the time constant is:

$$\tau_{e,b} = 1.298 - 1.25 = 0.048 \text{ s}$$

1. How do gain and time constant affect the system response?

The gain determines the magnitude of the system's output for a given input, while the time constant determines how quickly the system reaches its steady-state response.

2. Provide two reasons why the nominal model does not represent the SRV02 with better accuracy.

- **Parameter uncertainty:** The parameter values taken from reference data contain inherent uncertainty, so using them directly can introduce error into the model predictions.
- **Unmodeled dynamics:** To keep the experiment simple and avoid any unwanted complexities, several assumptions were made, which might lead to discrepancies between the model and the actual system behavior.

3. Explain how well the nominal model and the model obtained from the bump test model represent the SRV02 system.

The nominal model captures the overall first-order characteristics of the system but may deviate due to uncertainty in parameter values. The bump test model, since it is derived from experimental data, provides a closer match to the real system response. Although it is still affected by measurement noise, it represents the system more accurately than the nominal model.

Conclusion

This lab's objective was to create a transfer function model of the SRV02 servo system and test it experimentally to verify it. By supplying a step input voltage to the motor during the bump test, the steady-state gain and time constant can be obtained from the recorded data. After that, the model was assessed by changing these parameters in order to examine how they affected system behavior and to improve the model's fit to the real system response.

The results demonstrated that the time constant determines the system's response time, while the gain controls the output displacement's amplitude. Due to uncertainties in the assumed parameters, the nominal model did not accurately depict the physical SRV02 even if it represented the anticipated first-order dynamics. Because it was based on experimental observations, the model developed from the bump test offered a closer approximation within the measured operating range.

This experiment's main conclusion is that nonlinear effects cannot be well captured by a first-order approximation. Overall, the lab effectively illustrated the value of parameter modification in bringing simulated findings into line with actual system behavior and showed how experimental data can be leveraged to increase model accuracy.

References

[1] Moncayo H. Ph.D. Student Workbook Experiment 01. Embry-Riddle Aeronautical University. 2023.

[2] Quanser Inc. SRV02 User Manual, 2011.

Appendix A

The following shows the derivation for the nominal values, starting with equation 26 from [1]:

$$\omega_l(t) \cdot J_{eq} + B_{eq,v} \dot{\omega}_l(t) = A_m V_m(t)$$

$$\mathcal{L}\{\omega_l(t) \cdot J_{eq} + B_{eq,v} \dot{\omega}_l(t)\} = \mathcal{L}\{A_m V_m(t)\}$$

$$\Omega(s) \cdot s \cdot J_{eq} + B_{eq,v} \cdot \Omega(s) = A_m V_m(s)$$

$$\frac{\Omega(s)}{V_m(s)} = \frac{\frac{A_m}{B_{eq,v}}}{\frac{J_{eq}}{B_{eq,v}} \cdot s + 1}$$

$$K = \frac{A_m}{B_{eq,v}}$$

$$\tau = \frac{J_{eq}}{B_{eq,v}}$$

$$A_m = \frac{\eta_g K^2 \eta_m k_t}{R_m} = 0.128 \frac{N \cdot m}{V}$$

$$B_{eq} = \frac{\eta_g K_g^2 \eta_m k_t k_m + B_{eq} R_m}{R_m} = 0.084 \frac{N \cdot m \cdot s}{rad}$$

$$J_g = \frac{m_d r_D^2}{2} = 0.00005 \text{ kg} \cdot \text{m}^2$$

$$J_{eq} = \eta_g K_g^2 (J_{tach} + J_{m,rotor}) + (J_g + J_{l,ext}) = 2.15 \cdot 10^{-3} \text{ kg} \cdot \text{m}^2$$

$$K = \frac{A_m}{B_{eq,v}} = \frac{0.128}{0.084} = 1.53 \frac{(\frac{rad}{s})}{V}$$

$$\tau = \frac{J_{eq}}{B_{eq,v}} = \frac{2.15 \cdot 10^{-3}}{0.084} = 0.0253 \text{ s}$$

Appendix B

%Author: Anuranan Bharadwaj
% AE 443 - Lab 1

```
figure();

subplot(2,1,1)
plot(Vm(:,1), Vm(:,2), 'r', 'LineWidth', 1.5);
grid on;
xlabel("Time (s)");
ylabel("Voltage (V)");

subplot(2,1,2)
plot(wl(:,1), wl(:,2), 'LineWidth', 1.5);
hold on
plot(wl(:,1), wl(:,3), 'LineWidth', 1.5);
grid on;
xlabel("Time (s)");
ylabel("Rotational Speed (rad/s)");
```